Presentation 5

• Links to presentation(s) and code(s) on GitHub

- Presentation Slides: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/d488ea0d532a4d84ed336685ae9f054613a994e5/Presentations/Time%20Trends%20analysis/TimeTrends_8Dec_Jasmine.pdf
- PowerBI Dashboard: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/47b4eecda7bc608cac633b8d9da135925bdce380/Dashboard%20link/DrillDownCallVolume_AllCallsData_Dec8_Muse.pdf
- Code: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/47b4eecda7bc608cac633b8d9da135925bdce380/Code/Time%20Trends%20analysis/DrillDownCallVolume_AllCallsData_Dec8_Muse.ipynb
- Recorded Presentation: https://drive.google.com/file/d/1vh-3qAdm3LUhmC9gWf0edrORibGobjtu/view?usp=sharing

• What did you do?

For the final presentation, I focused on finalizing my drill-down visualization that shows the call journey for each intake line in All Calls Data. I made several improvements based on the last version: First, Instead of categorizing all unknown numbers into the broad “Unknown” category, I keep the original numbers themselves in the “Number Description” column. This way, we can identify the individual unknown numbers that receive a large call volume and analyze how to distribute the resources accordingly. Second, I re-examined the calculation for the y-axis – which is supposed to be the number of calls per weekday – and recognized the problem with the current DAX function. Instead of dividing the aggregate number of calls by the total number of weekdays in the dataset, it calculates the aggregate number of weekdays for each intake line and excludes those during which an intake line doesn't receive any call. I corrected this error by dividing all intake lines with the total number of weekdays in the time range of the dataset.

• How does it help the project?

First of all, by keeping all unknown numbers as they are, we present Legal Aid with a more detailed visualization, where they could figure out the call journey for each number instead of facing the black box in the broad “Unknown” category. Moreover, this new approach also provides more information for our current analysis: we can see exactly where each intake line is being transferred to and see the breath of each transfer level. For example, for the Main Number, which receives the most inbound calls, we can see that it is first transferred to 24 numbers, and then further transferred to 363 numbers during the second transfer level, and then transferred to 24 numbers again for the third transfer level.

Besides, I identified some unknown numbers that play important roles in the overall call journey. For instance, a significant proportion of calls received by the Main Number were first transferred to 2303 (10 calls per weekday), and the calls received by 2303 were further transferred to 187 unique numbers. This indicates that 2303, even as an unknown number, serves an important role as a transfer hub, and it's important to understand the wide range of unique calls it results in. 13125068649 is also a popular second transfer number for the Main Number. It is being transferred from Community Legal Clinics, Criminal Records, and HIV Intake Voicemail.

• Issues faced (if any)

I encountered a significant discrepancy in the Call Journey analysis where I was initially unable to replicate a specific high-volume call path identified by another team member, Tatum. While the reference analysis showed approximately 3,500 calls following a "Main Line → Front Desk → Main Line" journey, my analysis consistently returned zero results for this sequence.

Upon investigation, I discovered that this issue stemmed from subtle differences in how the raw Call Detail Records (CDRs) were processed. My initial approach filtered out calls with a duration of zero seconds, assuming them to be failed calls, and sorted the call legs based on "Start Time." These choices inadvertently deleted or scrambled the "Main Line" IVR steps, which are often instantaneous (zero duration) and share the exact same start second as the subsequent transfer leg.

• Attempts to resolve issues (if any)

To resolve this, I first conducted a side-by-side comparison of my code against the reference notebook to pinpoint the processing divergences. I then refined my data filters by removing the exclusion of zero-duration calls, ensuring that instantaneous transfers handled by the IVR are preserved rather than being treated as "phantom" calls.

Additionally, I implemented a more robust sorting mechanism by calculating an explicit "End Time" based on the system's "Release Time" and sorting the data by Correlation ID, End Time, and Start Time; this change ensured that the first leg of a transfer is correctly sequenced before the second leg begins.

Despite aligning the processing logic, the specific "return to Main" journey remained absent from the Inbound dataset. This led me to identify that the "return leg" from the Front Desk back to the Main Line is likely classified as an "Internal" or

"Outbound" call type by the vendor, causing these specific legs to be filtered out early in the pipeline due to the strict "Inbound" scope of my current analysis.

• Issues resolved (if any)

• Next steps

• References (Mention if you built up on someone else's work)

Presentation 4

• Links to presentation(s) and code(s) on GitHub
  - Presentation Slides: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/337db6d8e9f4a9c9efcd473363ac7b86ed35e48e/Presentations/Time%20Trends%20analysis/TimeTrends_19Nov_Muse.pdf
  - PowerBI Dashboard: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/bd0af2e9ad8b3925a80d773accc1a6832da7ba87/Dashboard%20link/DrillDownCallVolume_AllCallsData_Nov18_Muse.pdf
  - Code: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/bd0af2e9ad8b3925a80d773accc1a6832da7ba87/Code/Time%20Trends%20analysis/DrillDownCallVolume_AllCallsData_Nov18_Muse.ipynb
  - Recorded Presentation: https://drive.google.com/file/d/1uae4HHpAH9x-13Vp3RlFkvhPWEmdkuam/view?usp=sharing

• What did you do?

This week, I continued my detailed analysis of the AllCallsData dataset, specifically focusing on the called numbers, and implemented several improvements based on last week's feedback. My primary effort involved refining the call flow mapping to accurately indicate where inbound calls terminate. I processed the original data so that each row represents the complete flow for a specific call, where Number Description_1 is the intake number, Number Description_2 is the first transfer number, and so on. A significant addition this week was the introduction of an "ended" category for each sequential step. For example, if a call did not get transferred past the first step, Number Description_2 would show 'Ended.' This allows us to precisely evaluate the efficiency of each call by visualizing the proportion of calls that are terminated versus those that are transferred at every single step.

In addition to enhancing the termination status, I corrected an accuracy issue in the initial data handling by removing duplicated legs for the same transfer numbers. Last week, I had incorrectly counted both the originating and terminating record for each transfer as a unique leg. I have now deduplicated these records for each Correlation ID, ensuring the resulting call flow is accurate and truly reflects the sequence of unique connections.

Finally, a major deliverable this week was the development of an integrated visualization solution. I added a drill-down feature that allows users to explore and visualize all call flows through just one interactive chart. This capability simplifies the analysis of routing complexity and the overall organizational transfer network.

• How does it help the project?

This analysis is instrumental to the project because it provides actionable data to drive efficiency and strategic decision-making regarding call routing and resources. The core findings directly support two critical recommendations: rationalizing low-volume numbers and investigating the high volume of 'Unknown' calls.

The analysis of Call Flow Depth and Call Flow Width helps the project by quantifying the complexity of the current system, showing that routing complexity is highly centralized in the Main Number (which handles nearly 50% of calls requiring two or more connections), while simultaneously identifying many specialized lines with minimal transfers that could potentially be consolidated.

Furthermore, the developed drill-down feature gives the project a powerful, interactive tool to track the full lifecycle of every call, allowing stakeholders to trace individual call paths and view the aggregate distribution of transfers across the entire organization, which is crucial for future audits and optimization efforts.

• Issues faced (if any)

- It takes an extremely long time for the visualization to display

• Attempts to resolve issues (if any)

• Issues resolved (if any)

• Next steps

- Explore the services provided the unknown numbers

• References (Mention if you built up on someone else’s work)

Presentation 3

• Links to presentation(s) and code(s) on GitHub

- Presentation: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/655f3b0d797f54204bd5ff7f0fe9deae70d4993b/Presentations/Time%20Trends%20analysis/TimeTrends_6Nov_Jasmine.pdf
- PowerBI Dashboard: https://app.powerbi.com/groups/5a1f7ab1-8652-4d95-9529-92175c1572d2/reports/761ab1e1-885c-4388-a8fc-625b2b2244b5?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare
- Code: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/6a26c905ddc32ec34669d55bd09b740bb444ee1b/Code/Time%20Trends%20analysis/AddCallDescription%20%26%20MapOutCallLegsAsColumns_Nov5_Muse_Miao.ipynb
- Recorded Presentation: https://drive.google.com/file/d/1AKBUAvD_eqV614_SdD8fM_2jPnRUXEz_/view?usp=drive_link

• What did you do?

For this presentation, I visualized the call volume of inbound calls for both intake numbers and transfer numbers. To do this, I preprocessed All Calls Data to include only the inbound calls, and added a column called “Number Description” which details the services of the corresponding “Called number”. Since there are over 300 unique called numbers in the dataset but only around 20 have a clear description, I categorized the ones without description as “Unknown” so they are still included in the visualization.

In addition, I reorganized the dataset where I mapped out the call legs horizontally as columns “Number Description_1”, “Number Description_2”, “Number Description_3”, and “Number Description_4”, which respectively corresponds to the intake number, the first transfer number, the second transfer number and the fourth transfer number. Only the four numbers are kept since most calls ended within 3 transfers (with only one outlier having 25 transfers).

Finally, after processing the dataset, I developed visualizations to show call volume trends across different numbers for both intake and transfers. I also created DAX measures to calculate distinct correlation IDs per day, filtering for weekdays to standardize the y-axis for call volume.

• How does it help the project?

First, this preprocessing and visualization work makes the dataset more structured and interpretable, enabling stakeholders to understand the demand for different legal services without being overwhelmed by the raw numbers. Mapping the call sequences to descriptive labels rather than phone numbers provides actionable insights, such as identifying common call paths or bottlenecks.

Moreover, by mapping out the call legs horizontally as columns, we can better move between the transfers for each unique call and clearly understand how transfers are made between different numbers. By creating a drill-down feature, we can even choose one intake number and see how the calls are transferred to various services.

Lastly, standardizing the visualizations ensures that trends across different call legs are comparable, improving decision-making and operational analysis.

• Issues faced (if any)

At first, I had some difficulties using the DAX function to standardize the y-axis as the result of the DAX function kept showing zero. With multiple trials and the help of ChatGPT, I realized that the problem occurs because PowerBI cannot convert the date-time column “Start time” directly into date. By manually converting it to a date column, the DAX function works properly and can reflect the standardized value of call volume.

• Attempts to resolve issues (if any)

• Issues resolved (if any)

• Next steps

Based on my current visualization, I realized that Unknown numbers consist of most of the first transfer numbers. This highlights the need to understand their purposes, which would improve insight into overall call flows. In addition, based on the number description, the internal voicemail is not client related, but it’s still one of the most popular first and second transfer numbers for inbound calls in the All Calls Data. Therefore, it’s crucial to understand how this number is used for inbound calls.

• References (Mention if you built up on someone else’s work)

Work: Krish’s DAX function, which calculates the number of unique call per weekday

Purpose: standardize the y-axis for call volume visualization

Presentation 2

• Links to presentation(s) and code(s) on GitHub

- Presentation: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/13e1b70f16bbe55a94eb4a6bbac8aa17fbf063ba/Presentations/Time%20Trends%20analysis/TimeTrends_21Oct_Jasmine.pdf
- PowerBI Dashboard: https://app.powerbi.com/view?r=eyJrIjoiZDMyNzc4ZjQtMmE3Ny00YjVjLWJlOWYtNzY3Yjg2YzJkYzgzIiwidCI6IjdkNzZkMzYxLTgyNzctNDcwOC1hNDc3LTY0ZTgzNjZjZDFiYyIsImMiOjN9
- Code: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/ExtractingCallsWithExplainedNumbers_Oct21_Muse_Miao.ipynb

• What did you do?

For this presentation, I analyzed the volume of incoming calls for the intake lines with a clear explanation of their services (provided in the supplementary documents). While there are over 300 unique intake lines for incoming calls in All Calls Data, we only understand the purpose of around 20 lines. Therefore, I only kept the calls that were called to or from the list of the 20 intake lines, and visualized the call volume for each line, both aggregated and across time. For the time-series trend, I created two charts that show the call volume for the intake lines across hours in a day and across days in a week.

• How does it help the project?

In the document “Questions about Legal Aid Chicago Data”, it explicitly states that Legal Aid is interested in learning the call volume for each intake line. By visualizing the call volume by intake lines and time, we can better understand callers’ demand for each service, thereby knowing where or when to increase their staffing or to merge several intake lines together due to lack of demand.

Through the visualization of the aggregate call volume, I identified the 5 most popular lines (with over 10k calls received) as well as the 6 least popular lines (with less than 1000 calls received). I noticed that the Main Number and the Community Legal Clinics are called the most, and two internal routing numbers – which are “For those who already have an appointment” and “Transfer to English menu” – also appear pretty often. Then, “Farmworker main number/Migrant Legal Assistance Program” ranks fifth. The six least popular lines are for Trafficking Survivors Assistance Program, Fair Housing Intake voice mail, CLASP (Free Legal Help for Sexual Violence Survivors), Order of Protection Appeals Project (Help for Domestic Violence Survivors), A2J Immigration (Illinois Access to Justice, immigrate education and legal services), and Education Law Referrals voice mail. Within the past one and a half years, those lines received less than 1000 calls, which shows the potential of being combined or merged with other lines. My suggestion is to combine the intake lines for trafficking survivors, sexual violence survivors and domestic survivor into one line due to similar services, and to combine the intake line for A2J Immigration with the line for “Farmworker main number/Migrant Legal Assistance Program”.

Moreover, the time-series trends for the call volume also align with the aggregate pattern. From Monday to Friday, the proportion of call volume for each intake line remained relatively constant, with Main Number and Community Legal Clinics receiving the most calls. The total call volume reached a maximum on Monday and Tuesday, followed by a gradual decrease from Wednesday to Friday, and then a huge drop at weekends. In terms of hours in a day, the call volume reached a maximum between 9AM and 12PM, followed by a gradual decrease and a huge drop between 4-5PM. Besides, the proportion of call volume stays stable for the intake lines throughout the day; the number of incoming calls for Main Number began to decrease from 12PM, but the number of calls for Community Legal Clinics is relatively the same from 9AM to 3PM.

• Issues faced (if any)

I initially planned to visualize the average duration of each called number after learning about their call volume, but after taking a closer look at the “Duration” column in All Calls Data, I realized that the data input shows lots of inconsistency and cannot be used to compute the average call duration. Based on our current understanding, we think that the duration is recorded separately for each unique “Correlation ID” and “Called number”; however, I noticed that for most calls with the same “Correlation ID”, their legs show the same duration even when they were transferred to another number (indicated by “Called number” column). In addition, the duration of some calls is even shorter than the difference between “Start time” for different legs. This made me stop visualizing the average duration for now before fully understanding and cleaning the “Duration” column.

• Attempts to resolve issues (if any)

• Issues resolved (if any)

• Next steps

For my next steps, I plan to map the call flows and transfer patterns by linking internal routing numbers to their corresponding intake lines. Rather than treating these routing numbers as separate entities in my current visualization, I aim to integrate them into the overall workflow of incoming calls to show how callers move through the system. I will also explore whether transfers occur between multiple intake lines—for instance, whether a caller who reaches the main number can be redirected to the Community Legal Clinics or other services. This analysis will help identify common transfer paths, key entry points, and potential inefficiencies in the routing process that could be improved through streamlining.

Additionally, it is important to clarify the structure of outbound calls. Since there is no explicit "calling number" column, further investigation is needed to understand how outbound calls are logged. Fields such as "User number" may help indicate the source of these calls, allowing for a clearer picture of overall communication patterns. By combining inbound and outbound call volumes for each intake line, we can develop a more comprehensive understanding of the workload and communication flow within Legal Aid.

• References (Mention if you built up on someone else's work)

Presentation 1

• Links to presentation(s) and code(s) on GitHub

- Presentation: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_9Oct_Jasmine.pdf
- PowerBI Dashboard: https://app.powerbi.com/view?r=eyJrIjoiZDMyNzc4ZjQtMmE3Ny00YjVjLWJlOWYtNzY3Yjg2YzJkYzgzIiwidCI6IjdkNzZkMzYxLTgyNzctNDcwOC1hNDc3LTY0ZTgzNjZjZDFiYyIsImMiOjN9

- Code: https://colab.research.google.com/drive/1ey2SKEc4TqeqscitI0wzuhuyI-aX3NPp?usp=sharing

• What did you do?
- Explored the average call duration using AllCallsData, visualizing how the durations of both inbound and outbound calls vary throughout hours in a day and throughout days in a week.
- Proposed two metrics:
    - **Abandonment Rates**: using the “Answered” column from AllCallsData, the formula should look like Calls with “Answered = False” / total inbound calls. This new metric will help quantify missed opportunities for service and helps justify resource increases or chatbot automation. By looking at how the abandonment rates fluctuate over different time periods, we can discover when there’s a lack in staffing for LegalAid.
    - **Repeat Caller Ratio**: using the “Calling number” column from AllCallsData, we can identify the number of calling numbers with more than 1 inbound call / Total calls, which can help us identify if users are not getting help during their first calls. Ideally, the smaller the repeat caller ratio, the better LegalAid serves its callers.

• How does it help the project?

The average call duration in a particular time period is an important metric for improving LegalAid’s phone call system, since it provides an indirect measure of the complexity or the difficulty of the tasks handled by the calls. By looking at how the average call duration changes over time, we can understand both LegalAid’s operational schedule and callers' behavior. For example, if the average call duration for inbound calls in the afternoon is longer than that in the morning, it indicates that callers tend to inquire about more complex issues in the afternoon. Similarly, if the average call duration for outbound calls in work days is longer than that at weekends, it could suggest that LegalAid staff tend to tackle more time-consuming issues through Monday to Friday. We can discover whether there’s misalignment between the organization’s schedule and callers’ expectations by examining the average call duration of inbound and outbound calls.

Based on my visualization for the average call duration throughout hours in a day, I found that the average duration for both inbound and outbound calls stay relatively similar from 5AM to 5PM. However, after 5PM, the average duration of outbound calls reduced significantly, while that of inbound calls increased dramatically and reached its peak between 8PM and 10PM. This divergence indicates that, while the call operators might be winding down their operations and tend to address simpler issues after normal work hours, the callers are actually calling to deal with more complex or more time-consuming tasks. The pattern of average call duration suggests a possible misalignment between the organization's operational hours and the public's needs, implying that resource allocation or staffing during evening hours may not fully align with when callers expect support for more time-consuming issues.

For the visualization of average call duration throughout days in a week, both the inbound and outbound duration remains relatively the same from Tuesday to Sunday, yet the duration of inbound calls is about 1-minute longer than the duration of outbound calls on Mondays, which might suggest that the callers tend to inquire about more time-consuming issues at the start of a week yet the call operators prefer to handle simpler issues reflected by the shorter duration of outbound calls.

To better align LegalAid's operations with caller needs, the organization could consider extending or reallocating staff coverage during evening hours, when callers tend to bring up more complex issues. Additionally, scheduling more experienced staff or specialized training sessions earlier in the week—particularly on Mondays—may help address the surge in longer inbound calls. Regularly monitoring call duration patterns by time of day and day of week can further guide data-driven adjustments in staffing and scheduling to improve overall service efficiency and caller satisfaction.

• Issues faced (if any)
While I initially used the "Direction" column to determine if a call is inbound or outbound, this column alone cannot show if the calls are truly inbound or outbound, since it only indicates whether a call originates or terminates at LegalAid. I need to also take the "PSTN vendor name" column into consideration, looking at the values for both columns and removing the internal calls to avoid confusion.

• Attempts to resolve issues (if any)
To distinguish between inbound and outbound calls, I first sorted the observations in AllCallsData by "Correlation ID" and "Start time," then retained the first record for each unique correlation ID. Calls were categorized based on the "PSTN vendor name" and "Direction" fields: if the "PSTN vendor name" was "CallTower" and the "Direction" was "Terminating," the call was classified as inbound; if "CallTower" and "Originating," it was classified as outbound. When "PSTN vendor name" was "NA," and "Direction" indicated either "Terminating" or "Originating," the call was identified as internal. Following this classification, I aggregated the "Duration" values for all rows sharing the same correlation ID to accurately represent the total duration of each call before re-visualizing the average call duration.

• Issues resolved (if any)

By preprocessing the data through methods mentioned above, I was able to correctly visualize the average call duration for inbound and outbound calls.

• Next steps

Assuming that the inbound and outbound calls have been correctly identified, I plan to investigate average call duration in the following directions:

- Identify if specific phone menus or call/client types are associated with the time periods showing higher average call duration
- Examine caller wait times during periods with higher average call durations to assess potential understaffing during peak hours
- Evaluate whether call timing influences case outcomes
- Assess if periods with long average call duration are seasonal or constantly recurring

• References (Mention if you built up on someone else’s work)